A Literature Review of "Online Images Amplify Gender Bias" (Guilbeault et al., *Nature*, 2024)

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***Abstract*—This paper is a literature review, not a report of original research. It summarizes and reflects on Guilbeault et al.'s 2024 *Nature* study, "Online images amplify gender bias," as synthesized for a CS 5002 (Discrete Structures) research-review presentation at Northeastern University in Spring 2024. The reviewed study argues that, as online content shifts from text to images, gender bias embedded in visual media is stronger than the corresponding bias in text, and that human gender associations track image-based bias more closely than text-based bias. This review restates the reviewed paper's methodology and key findings as they were presented in the original 15-slide class deck, and discusses their implications for AI/ML fairness work. No new data, experiments, or code were produced as part of this review; all empirical claims below are attributable to the original authors.**

I. Introduction

Machine learning fairness research has historically focused on bias in text corpora and text-derived word embeddings, since text has been the dominant medium for large-scale computational analysis. However, the modern internet is increasingly dominated by images: search engines, social media feeds, and recommendation systems surface visual content at a scale that rivals or exceeds text. Guilbeault et al. [1] investigate whether gender bias in this visual channel is comparable to, weaker than, or stronger than bias in text, and whether the two channels differentially shape human beliefs.

This paper does not present new experiments, code, or data. It is a literature-review synthesis, originally produced as a 15-slide class presentation for CS 5002 (Discrete Structures) at Northeastern University, Spring 2024, by a five-person student group (Yao Chong Chow, Kaustubha Eluri, Yuneng Li, Jonathan Mowat, and Nicholas Ung). The assignment required reading a peer-reviewed paper, synthesizing its methodology and findings, and presenting implications to a classroom audience — not reproducing the underlying experimental work. This write-up restates that synthesis in IEEE conference-paper form, with full credit to the original authors for all empirical content.

II. Summary of Reviewed Work

The reviewed paper, Guilbeault, Delecourt, Hull, Desikan, Chu, and Nadler (2024), "Online images amplify gender bias," published in *Nature*, presents a large-scale multimodal study spanning over one million images and a controlled experiment on a nationally representative human-participant sample. The following subsections summarize the paper's methodology and results as covered in the reviewed slide deck.

A. Image Collection

The authors drew roughly 3,500 social categories (e.g., occupations) from WordNet and collected corresponding images by scraping Google Image Search and Wikipedia for each category.

B. Face Extraction

An OpenCV-based deep-learning face-detection pipeline was applied to each collected image to extract cropped faces. This step reduces confounding signal from background, posture, and clothing, isolating the gender-classification target to the face itself.

C. Human Gender Classification

Extracted faces were classified as male, female, or neither by Amazon Mechanical Turk (MTurk) workers. The authors computed inter-rater agreement statistics and recorded coder demographics to assess the robustness of the labeling process.

D. Gender Dimension in Word-Embedding Space

For the text side of the comparison, the authors used pretrained word-embedding models (e.g., GloVe) to compute a gender direction in embedding space, defined as the average cosine distance between each social category and predefined clusters of male- and female-associated words.

E. Human-Participant Experiment

A nationally representative panel was recruited through Prolific. Participants were randomized into three conditions — image-search, text-search, and control — and their gender associations were measured both explicitly, via Likert-scale ratings, and implicitly, via an Implicit Association Test (IAT).

F. Correlation Analysis

Bias scores derived from online images, online text, and human participant associations were correlated against one another. The central empirical result is that human gender associations track image-based bias more strongly than text-based bias.

G. Key Findings

As highlighted in the reviewed slides, the paper's central findings are:

- **Image bias exceeds text bias** for the same occupational categories. For example, the category "programmer" yields a near-neutral gender association (≈ 0) in text, but a stronger male-skewed association (≈ 0.3) in images.

- **Human associations fall between image and text bias**, but align more closely with image bias, suggesting that visual exposure plays an outsized role in shaping cultural gender associations relative to textual exposure.
- **Visual content amplifies and entrenches gender stereotypes** more than equivalent textual content does, even when drawn from comparable source categories.

III. Discussion

The reviewed paper's findings carry direct implications for AI and machine learning fairness research, which the class presentation discussed in its concluding slides.

First, bias-mitigation techniques that operate purely on text — such as debiasing word embeddings or filtering biased language in training corpora — may miss a substantial share of the bias that reaches end users, since the visual channel appears to carry a stronger and more influential bias signal. Systems built on image data, including search ranking, social media recommendation, and vision-language or text-to-image generative models (e.g., CLIP-style embeddings and diffusion-based image generators), may inherit and further entrench this stronger visual bias unless explicitly audited.

Second, the paper's methodology models a template for multimodal bias auditing: pairing large-scale automated content analysis (image scraping, face detection, embedding-space bias metrics) with a controlled human-subject experiment that links measured content bias to actual human belief formation. This two-sided design — bias-in-content and bias-in-people — is more convincing than either component alone, and is a pattern reproducible for other domains (e.g., race, age, or ability bias in images).

Third, the paper explicitly flags directions for future work that the class discussion echoed: extending the analysis to demographic dimensions beyond gender, comparing human-generated versus AI-generated visual content, and running intervention studies to test whether reducing image-level bias measurably shifts human associations over time. These directions are increasingly relevant as generative AI models produce a growing share of the images that populate the internet, potentially compounding the bias-amplification effect the authors document for human-curated content.

It should be emphasized that all of the above is an interpretation of the original authors' contribution, produced for a classroom research-review exercise. No independent verification, reanalysis, or extension of the underlying data or methods was performed by the author of this paper.

IV. Conclusion

This paper has summarized, in IEEE conference format, a literature-review presentation on Guilbeault et al.'s 2024 *Nature* study of gender bias in online images versus text. The reviewed study finds that gender bias is stronger and more influential in images than in text, and that human gender associations track visual bias more closely than textual bias, with implications for AI/ML fairness work that has historically prioritized text-based bias audits. This document is a synthesis and reflection produced as coursework for CS 5002 at Northeastern University, not a report of original research, data collection, or experimentation.

References

[1] D. Guilbeault, S. Delecourt, T. Hull, B. S. Desikan, M. Chu, and E. Nadler, "Online images amplify gender bias," *Nature*, 626, 1049–1055, 2024.